COMPUTER SYSTEMS AND PLATFORM TECHNOLOGIES

IT0047

FEU ALABANG FEU DILIMAN FEU TECH

Technology Driven by Innovation

MODULE 4

The System Bus, Von Neuman Architecture & CPU Registers

At the end of the lesson, the student should be able to :

- Define Bus
- Identify types of Bus
- Identify the Memory Operations

MODULE 4.1

The Bus System

At the end of the lesson, the student should be able to :

- **Define Bus**
- **Identify types of Bus**
- **Identify the Memory Operations**

Buses

- Bus - A bus is a collection of wires on which electrical signals pass between components in the system.
- Data Bus - The 80x86 processors use the ***data bus to shuffle data between the various components in a computer system. The size of this bus varies widely in the 80x86 family.***

80x86 Processor	***Data Bus Size***
8088	8
80188	8
8086	16
80186	16
80286	16
80386sx	16
80386dx	32
80486	32
80586 / Pentium family	64

Address Bus

- Address Bus - The ***address bus specifies the location of data in memory. To differentiate memory locations and I/O devices, the system designer assigns a unique memory address to each memory element and I/O device***

- The number of bits on the address bus will determine the maximum number of addressable memory and I/O locations (2^n) n- bit binary number. The 8088 and 8086, for example, have 20 bit address busses. Therefore, they can access up to 1,048,576 memory locations.

80x86 Family Address Bus Sizes

Processor	***Address Bus*** Size	Max. Addressable	Memory In English
8088	20	1,048,576	1 Megabyte
8086	20	1,048,576	1 Megabyte
80188	20	1,048,576	1 Megabyte
80186	20	1,048,576	1 Megabyte
80286	24	16,777,216	16 Megabytes
80386sx	24	16,777,216	16 Megabytes
80386dx	32	4,294,976,296	4 Gigabytes
80486	32	4,294,976,296	4 Gigabytes
80586/Pentium	32	4,294,976,296	4 Gigabytes
Pentium Pro	36	68,719,476,736	64 Gigabytes
Pentium II	36	68,719,476,736	64 Gigabytes
Pentium III	36	68,719,476,736	64 Gigabytes

Control Bus

- **Control Bus** - The control bus is an extensive collection of signals that control how the processor communicates with the rest of the system.
- The ***read and write*** *control lines control the direction of data on the data bus. When both contain a logic one, the CPU and memory-I/O are not communicating with one another. If the read line is low (logic zero), the CPU is reading data from memory (that is, the system is transferring data from memory to the CPU). If the write line is low, the system transfers data from the CPU to memory.*
- The ***byte enable lines*** *are another set of important control lines. These control lines allow 16, 32, and 64 bit processors to deal with smaller chunks of data.*

Data Bus

The data bus is used to transfer data to and from the memory or I/O devices. It is bi-directional.

The address bus is used to select the desired memory or I/O devices by providing a unique address that corresponds to one of the memory or I/O. It is unidirectional.

The control bus is used to carry control signals to and from the memory or I/O devices. It is bi-directional.

Memory Operations

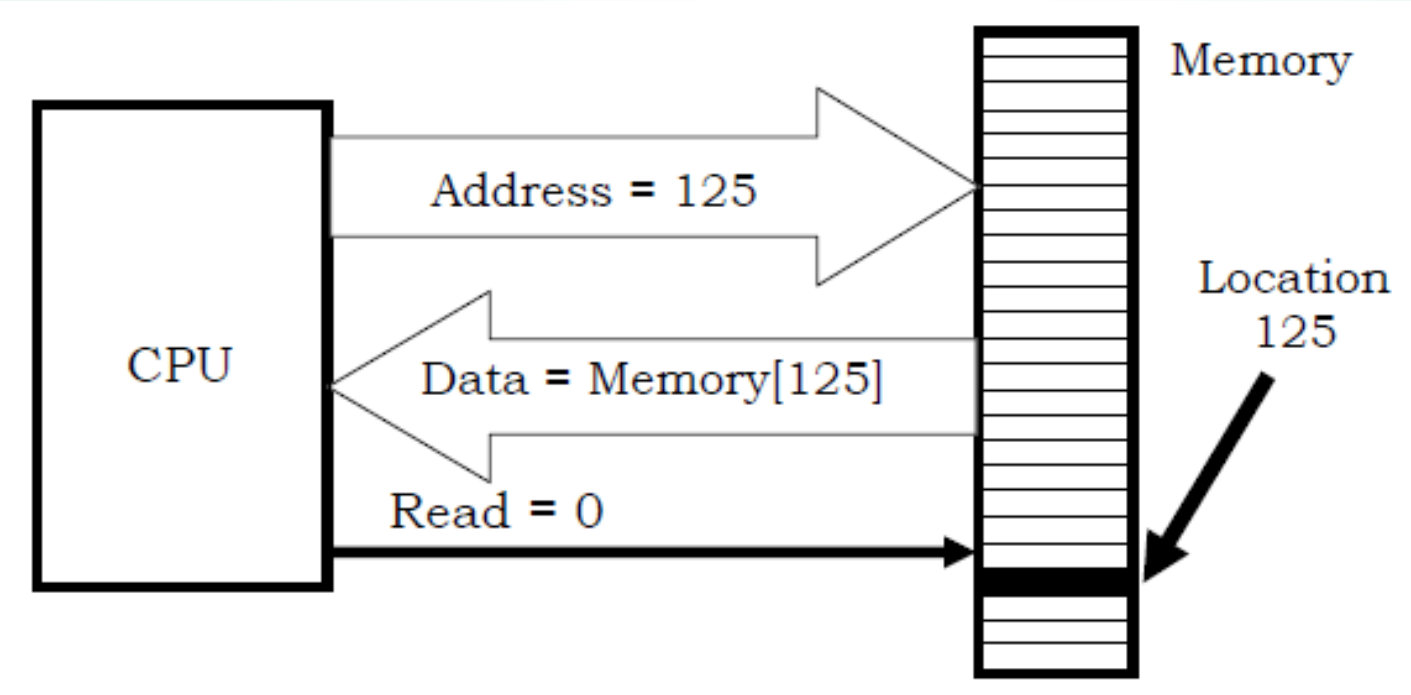
Fetch or Read. This transfers the contents of a specific MM location to the CPU. The word in the MM remains unchanged.

Read/Fetch Cycle:

1. CPU sends address of the desired location.
2. MM reads the data stored at that address and sends it to the CPU.

Reading the contents of Address 125.

The CPU places the address 125 on the address bus, enables the read line (since the CPU is reading data from memory), and then reads the resulting data from the data bus.



Memory Operations

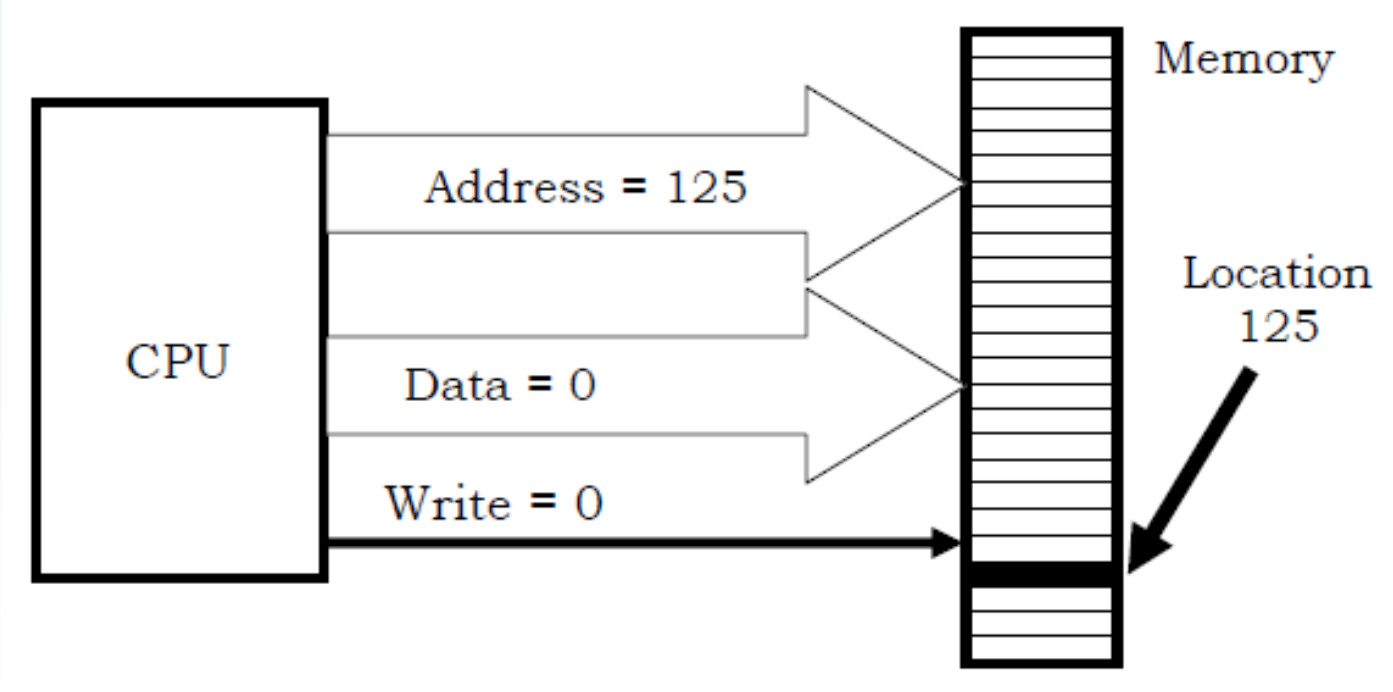
Store or Write. This transfers a word of information from the CPU to a specific MM location. This overwrites the former contents of that location.

Write/Store Cycle:

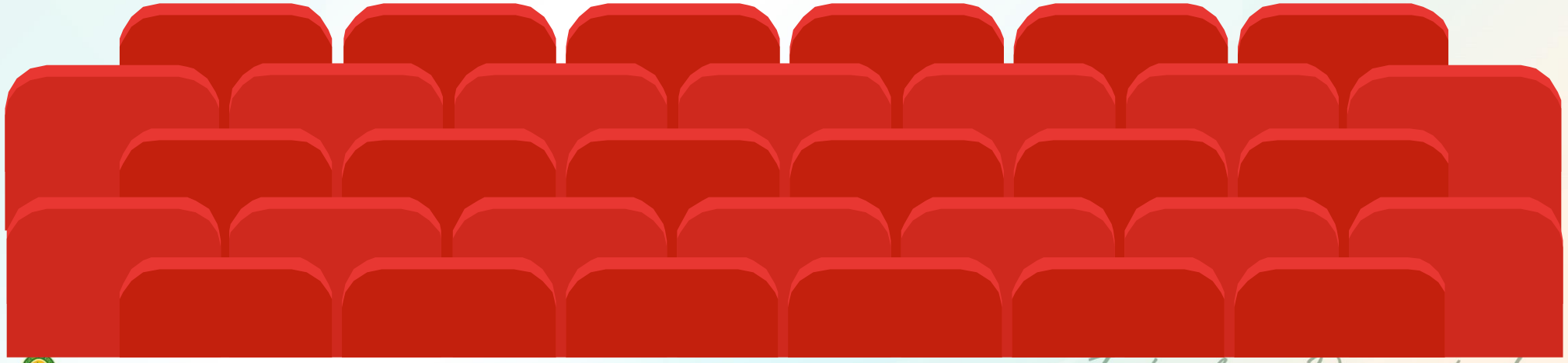
1. CPU sends address of the desired location to the MM, together with the data to be stored into that location.
2. Data is written at desired location.

Initializing the contents of address 125 to the value 0.

The CPU places the address 125 on the address bus, the value zero on the data bus, and enables the write line (since the CPU is writing data to memory).



Q&A SESSION

ASK ANY QUESTION RELATED TO OUR TOPIC FOR TODAY.

THANK YOU